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324356 (24)

BE (3rd Semester)

Examination, Nov.-Dec., 2014

Branch : Electrical Engineering

ELECTRICAL POWER GENERATION (NEW)

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Attempt all questions. In each question part 'a' is compulsory, from remaining attempt any two parts. All questions carry equal marks. Assume suitable data, missing if any.

Q. 1. (a) What is spinning reserve ? 2

(b) How does the load factor and diversity factor affect the cost of electrical energy ? 7

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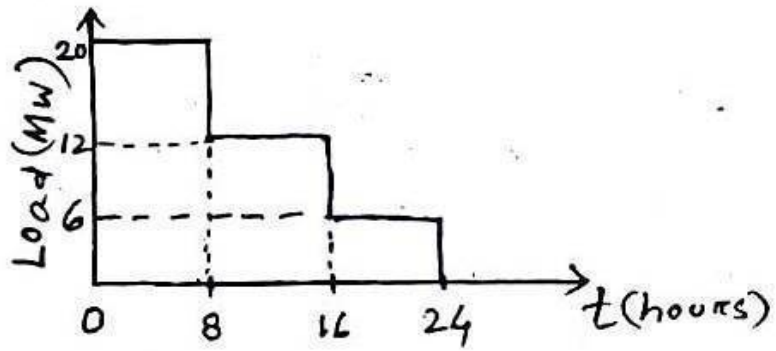
(c) (i) Define chronological load curve and load duration curve. 2

(ii) The load duration curve for a system is shown in fig. Plot energy load curve and find the load factor. 5



(d) Determine the generation cost per unit of energy from the following plant data. 7

Installed capacity = 120 MW

Capital cost of plant = Rs. 40,000/kW.

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Interest and depriciation = 15%

Fuel consumption = 0.64 kg/kWh.

Fuel cost = Rs. 1500/1000 kg.

Salaries, wages, repair and other operating cost per annum = Rs. 50,000,000.

Peak load = 100 MW

Load factor = 60%

Q. 2. (a) Draw the water-steam flow diagram of a steam power plant. 2

(b) Name and explain some of the categories of heat losses in a thermal plant. Discuss the advantages and disadvantages of using pulverised coal in steam plant. 7

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(c) Write notes on : 3½×2=7

(i) Economiser

(ii) Air-preheater

(d) What are the main features of 9 m pulse and reaction turbines in a thermal power station ? 7

Q. 3. (a) Nuclear power plants are used for base load plants, why ? 2

(b) Define the following terms using suitable diagrams : 2+2+3

(i) Fore bay

(ii) Spill way

(iii) Pressure swing

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(c) Discuss the types of dam used in hydro-electric stations. 7

(d) Explain the working of pressurised water reactor with suitable diagram. 7

Q. 4. (a) What is solar radiation ? 2

(b) Discuss different types of solar collectors. What are the advantages of concentrating collector over a flat plate collector ? 7

(c) Write notes on : 3½×2

(i) Savonius rotor

(ii) Wind energy storage

(d) Describe a central receiver system for a central solar thermal power station. 7

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Q. 5. (a) What is a hybrid MHD system ? 2

(b) With neat sketch draw a open cycle MHD system and explain its working. 7

(c) Write notes on Tidal power generation. 7

(d) What are the possible sources of Geothermal pollution ? How these can be minimised ? 7

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